21 Vertebrate evolution

Before you attempt this section, remind yourself of the following classification terms.

Class, Family, Genus, Kingdom, Phylum, Order, Species

1 Rewrite them in order, from the least specific to the most specific.



Many species in the phylum *Chordata* are considered the most advanced of animals. It was long believed that the chordates evolved from the *Echinoderms*, meaning 'spiny skinned animal' (eg starfish). However, most scientists now believe the two groups *co-evolved* from a different, common ancestor. Since most of the early chordates were soft-bodied and believed to have arisen about 700 million years ago (during the Pre-Cambrian era), it is near impossible to find fossil evidence to confirm what embryonic development patterns suggest about their relationship.

2 What does *co-evolve* mean?

A distinguishing feature that separates the first vertebrates from the chordates is the presence of a *notochord* at some point during development. The notochord is a collagen-based rod that strengthens the dorsal (back) part of an animal and develops into a vertebral column (backbone). Collagen is a fibrous protein that can make up connective tissue. The notochord commonly extends from the tail to the head and, as the animal develops, it is replaced with the vertebral column. The original chordates were believed to be slender, soft-bodied, filter-feeding animals. The earliest chordates were the ancestors of the vertebrates. The first vertebrate form was the *Agnathans*, or jawless fishes. The hagfish and lampreys are examples of jawless fishes but, interestingly, the two groups did not evolve from the same ancestor.

3 Lampreys exist on the blood of other fishes. Which of the following could not be used to define lampreys according to the description of Agnathans?

A Parasites **B** Marine **C** Carnivore **D** Invertebrate

The phylum Chordata evolved into three branches:

- *Urochordata:* These are often found in colonies and are commonly sessile (not free to move about). Examples include tunicates (these are covered in a protective 'tunic'), sea squirts, cunjevoi, sea cucumber.
- *Cephalochordata:* This 'middle' group is in many ways similar to the vertebrates in having evolved a circulatory system. However, this species do not have a true heart, for example amphioxus.
- *Vertebrata:* This most advanced group possesses a heart, brain, circulatory system and vertebral column, for example humans.

Radial symmetry is when there are similar parts regularly arranged around a central point. Vertebrates are bilaterally symmetrical.

4 Give a definition for *bilateral symmetry*.

5 Compare *bilateral symmetry* with *radial symmetry*.

6 To which phylum do we belong? We include sea urchins, brittle stars and sea cucumbers. We have an internal skeleton, often have five parts, are marine animals and sometimes have spikes.

7 Give a reason why there are few fossils of early chordates.

Research

8 Jean Lamarck (1744–1829) came up with a theory of evolution 50 years before Charles Darwin. Give a brief summary comparing Lamarck's theory of evolution with Darwin's.

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